25901402

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEMBG200 - INDUSTRIAL MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **1** mark.

1. What is the scientific name of baker's yeast? [K] / [CO1]

   Saccharomyces cerevisiae -1 mark

2. Differentiate continuous fermentation from batch fermentation based on nutrient supply. [K] / [CO2]

   Continuous fermentation: Nutrients are continuously supplied- ½ mark

   Batch fermentation: Nutrients are added only at the beginning- ½ mark

3. What is crystallisation? [K] / [CO3]

   Used to obtain pure solid products (e.g., antibiotics, organic acids, sugars) - 1mark

4. What is adsorption? [K] / [CO3]

   Attachment of enzymes or cells onto a solid support surface by weak physical forces. 1 mark

5. Name one physical method used for industrial sterilisation. [K] / [CO2]

Heat -e.g., Autoclaving, Filtration, Radiation

Any one correct method → 1 mark

[1x5 = 5]

Part B

Short Answer Questions

Answer any **10** questions.

Each question carries **2** marks.

6. If contamination is detected after reviving a preserved culture, what steps would you apply to prevent recurrence? [A] / [CO1]

   Identify and eliminate contamination source-1 mark, Apply strict aseptic techniques- 1 mark.

7. Analyse how dual fermentation improves product yield. [An] / [CO2]

   Utilises two stages/microorganisms or pathways efficiently – 1 mark

   Enhances substrate utilisation and increases overall yield/productivity – 1 mark

8. Name two materials used for entrapment. [K] / [CO3]

   Any two correct examples (e.g., alginate, polyacrylamide, agar) – 1 + 1 = 2 marks.

9. Explain why penicillin fermentation is carried out under aerobic conditions. [U] / [CO4]

   Fungi require oxygen for growth/metabolism – 1 mark

   Oxygen is essential for the biosynthesis of penicillin – 1 mark

10. Which microorganism is used in penicillin production, and in which industry is it applied? [A] / [CO1]

    Penicillium chrysogenum (or *P. notatum*) – 1 mark

    Pharmaceutical industry – 1 mark

11. Explain the importance of trace elements in fermentation media. [U] / [CO1]

    Act as cofactors/activators for enzymes – 1 mark

    Support microbial growth and metabolic activities – 1 mark

12. Explain why batch fermentation is considered a closed system. [U] / [CO2]

    No addition of nutrients after inoculation – 1 mark

    No removal of products/waste until completion – 1 mark

13. What is the role of carriers in immobilisation? [U] / [CO3]

    Provide support matrix for cells/enzymes – 1 mark

    Enhance stability, reusability, and activity – 1 mark

14. Explain the effect of temperature on citric acid fermentation. [U] / [CO4]

Optimum temperature required for maximum production – 1 mark

High or low temperature reduces enzyme activity and yield – 1 mark

15. Identify why testing in flasks helps in secondary screening. [K] / [CO1]

Provides conditions similar to industrial fermentation – 1 mark

Helps evaluate yield and productivity quantitatively – 1 mark

16. Identify two primary screening methods for antibiotic producers. [K] / [CO1]

Crowded plate technique – 1 mark

Agar diffusion (zone of inhibition) method – 1 mark

17. Examine why the accumulation of toxic by-products is reduced in fed-batch culture. [An] / [CO2]

Controlled addition of substrate prevents excess accumulation – 1 mark

Maintains optimal conditions, reducing inhibitory/toxic products – 1 mark

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions.

Each question carries **6** marks.

18. Discuss the various types of centrifuges used for downstream processing. [An] / [CO3]

Definition/principle of centrifugation – 1 mark

Explain any four types of centrifuges -4 marks

Applications/advantages in downstream processing – 1 mark

19. Explain the role of cell membrane permeability in glutamic acid secretion and the methods used to enhance its excretion from cells. [U] / [CO4]

Importance of membrane permeability in secretion – 2 marks

Mechanism of glutamic acid release -biotin limitation - membrane alteration – 2marks

Use of surfactants, antibiotics, etc.- 2 marks

20. Describe the contributions of Alexander Fleming to industrial microbiology. [A] / [CO1]

Any 3 important points- 3x2 =6

21. Discuss the desirable properties of industrial strains with suitable examples. [U] / [CO1]

Any 3 important characteristics with examples -3x2 =6

22. What is a fermenter? Write down its functions. [U] / [CO2]

Fermenter definition– 1 mark

Any 5 Functions - 5marks

23. Describe the role of immobilised enzymes in improving industrial product quality and purity. [K] / [CO3]

Definition of immobilised enzymes – 1 mark, Advantages -4 marks

Industrial examples/applications – 1 mark

24. Classify amylase based on the biochemical mechanism of starch hydrolysis. [U] / [CO4]

Amylase introduction – 1 mark, α-amylase– 1 mark, β-amylase enzyme– 1 mark
Differences in mode of action – 2 marks, Applications – 1 mark

[6x5 = 30]

Part D

Essay Questions

Answer any **1** question.

Each question carries **15** marks.

25. Explain the basic requirements and functions of an ideal fermenter in industrial microbiology. [U] / [CO2]

Introduction – 2 marks, Basic Requirements of an Ideal Fermenter -6 marks,

Functions of a Fermenter - 5 marks, Neat, labelled diagram – 2 marks

26. Explain covalent bonding and crosslinking methods in immobilisation. [An] / [CO3]

Definition of immobilisation -2 marks,

Covalent Bonding Method- Principle, mechanism, supports used, Advantages, Disadvantages- 5 marks

Crosslinking Method - Principle, mechanism, supports used, Advantages, Disadvantages- 5 marks

Diagram of covalent bonding and crosslinking – 2 marks, Applications- 1 mark.

[15x1 = 15]